

Problem 1
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 100. $\frac{1}{2}$

1. If flag pattern = JE w/in IP data $\rightarrow$ 7D 5E
If escape seq = 7D w/in IP data $\rightarrow$ 7D 5D

If IP data is indep binary seq w/ $\Pr\{1\} = \frac{1}{2}$, find percentage inc in avg data rate due to byte stuffing

1. input is scanned serially bit by bit, for when 7E or 7D occur ANYWHERE

prob of 7D = $\frac{1}{2^7} = \frac{1}{256} \hat{=}$ prob of 7B. is well

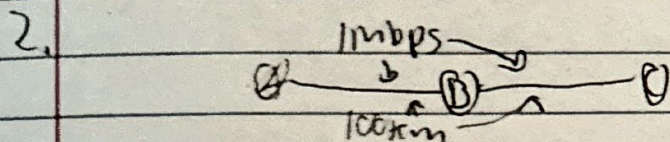
$$\frac{1}{256} \times 2 = \frac{1}{128} = .78125\% \text{ chance of 8 bits in, 50}$$
$$8(.18125) = \sqrt{6.2590} \text{ in } \text{mg} \cdot \text{denote rate}$$

ii Byte pattern only now. 0000 0111 1110 0000X/
0111 1110 0000 0000✓

some thing but no x8

.7812500

b. Max possible increase is 100%, and this occurs when all bytes are 7D or 7E or a mixture.



$$\frac{100,000}{2 \times 10^8} = 5 \times 10^{-4} \text{ s to travel}$$

1500 bytes message into 4
375 byte packets. Each packet
has additional 4 byte header
A send w/out ACK from B.

B must receive full packet
before sending. $c = 2 \times 10^8 \text{ m/s}$
How long until C has with message?

$$375 + 40 = 415$$

$$\frac{415 \times 8}{10^6} = 3.32 \times 10^{-3} \text{ s for A to send one message}$$

$$4(3.32 \times 10^{-3}) = 13.28 \times 10^{-3} \text{ s} + 5 \times 10^{-4} = 13.78 \times 10^{-3}$$

$$13.78 \times 10^{-3} + 3.32 \times 10^{-3} = 17.1 \times 10^{-3} + 5 \times 10^{-4} = 17.6 \times 10^{-3}$$

17.6 ms

56-kbps data stream using spread spectrum

3.

(a) Find the channel bandwidth required to achieve a 56-kbps channel capacity when $SNR = 0.1, 0.01, \text{ and } 0.001$

$$56000 = B \log_2(1 + 1) \rightarrow B = 407262.2403 \text{ Hz}$$

$$SNR = 0.01 \rightarrow B = 3901000.146 \text{ Hz}$$

$$SNR = 0.001 \rightarrow B = 38835647 \text{ Hz}$$

b. min SNR to get bandwidth = capacity w/out appreciable errors

$$1 = \log_2(1 + SNR)$$

$$2 = 1 + SNR$$

$1 = SNR \rightarrow$ much higher vs the spread spectrum (use (a))

4. a synch nonstat TDM is used to combine four 4.8-kbps and one 0.6 kbps signals for transmission over a single leased line. For framing, a block of 7 bits (pattern 1011101) is inserted for each 48 data bits, are framing bits?

1. arb select pos

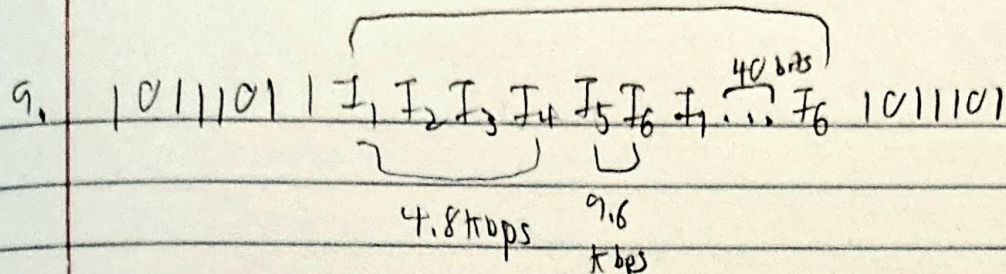
2. consider the block of 7 ctrl bits

3. observe first block of 7 bits each frame for 12 consecutive frames

4. if 10/12 blocks match, we in frame; else adv one bit pos

and go to step 2

48 bits



b. % overhead?

$$\frac{2}{48+2} = 12.72\%$$

c. multiplexed output bit rate

$$\frac{4800 \text{ bps}}{8 \text{ b}} (48+2) = 33 \text{ kbps}$$

d. frame times

min = first frame correct

$$\frac{12 \times 55}{33000} = 0.2 \text{ s}$$

max = last frame correct

$$\frac{12 \times 55}{33000} \times 55 = 1.1 \text{ s}$$

$$\text{avg} = \frac{1.1 + 0.2}{2} = 0.65 \text{ s}$$